Structural identifiability, Fisher geometry, and branch-aware inversion of a time-averaged ESPI Bessel operator

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August 15, 2026

Abstract

Time-averaged electronic speckle pattern interferometry (ESPI) maps a signed vibration-amplitude field to an even, oscillatory Bessel response, so quantitative inversion must confront global nonuniqueness and strongly nonuniform local sensitivity at once. We treat the calibrated second-order observable as a probabilistic field operator, $\mu(W) = A + BJ_0^2(cW)$, and characterize its inverse geometry. The central structural result concerns the signed-field inverse fiber: regionwise sign reversals generate up to $2^{N_{\mathrm{nodal}}}$ sensor-equivalent fields, but a connected regular-interface C^1 mechanical class contracts this set to exactly the global pair $\{W, -W\}$; a six-domain plate mode contracts 64 optical candidates to two. At bright and dark fringe extrema the squared-Bessel map instantiates the classical nonregular quarter-power regime: the measurement Jacobian vanishes while quadratic curvature remains, and the observed error scaling is near $N^{-1/4}$. The dependence on the products ca_i also exposes an exact scale orbit, so the Bessel channel identifies optical-phase products rather than metric amplitudes. Branch-aware maximum-likelihood studies verify these predictions, and a 1,000-realization full-field test shows that the modal covariance predicts the spatial uncertainty map $\Phi F_{\boldsymbol{a}}^{-1}\Phi^T$ with rank correlation 0.998 and near-nominal interval coverage. An externally phase-referenced vibration channel then supplies the missing sign and scale information through a single design quantity Q. A balanced 100,000-trial binary calibration confirms the closed-form error law $\Phi(-\sqrt{Q})$ to maximum absolute deviation 0.0012, while a separate 1,000-realization profile-MLE test verifies the same monotone design trend when the optical magnitude is estimated rather than treated as known. The framework separates optical symmetry, mechanical admissibility, local statistical conditioning, and auxiliary information within one branch-aware inverse problem.

1 Introduction

Full-field optical metrology provides dense measurements of mechanical vibration, while the recorded ESPI image remains an optical observation of the underlying displacement field. Quantitative inference therefore requires a forward map from the mechanical field to a specified optical observable and a statistical model for the fluctuations of that observable.

For harmonic out-of-plane motion, time averaging produces a zero-order Bessel modulation of the coherent interference term [1, 2]. Several established ESPI processing schemes subsequently form correlation, subtraction, contrast, or phase-derived observables [2, 3, 7]. The present work focuses on a second-order correlation/contrast statistic for which the deterministic mean is proportional to the squared magnitude of the time-averaged complex modulation. This choice yields

$$W \longmapsto p(I \mid \mu(W), \eta), \qquad \mu(W) = A + BJ_0^2(cW), \tag{1}$$

where W is a signed real modal-amplitude field referenced to a common temporal phase, A is the background level of the selected observable, B is its modulation coefficient, c is the optical displacement-sensitivity coefficient, and η collects statistical and acquisition nuisance parameters. The coefficients A, B, and μ have the same units; B is therefore a modulation-intensity coefficient rather than a dimensionless fringe visibility.

The inverse structure of Eq. (1) contains two distinct global ambiguities. Evenness makes the optical observation invariant to admissible sign changes of the modal field, while oscillation of J_0^2 creates multiple amplitude branches. Local inversion adds a separate issue: the measurement Jacobian vanishes at stationary bright and dark fringe extrema. These global and local effects require different remedies and lead to different uncertainty statements.

The analysis follows this geometry from the sensor map to the inverse problem. The field Jacobian and stationary curvature determine where local information is available. The nodal partition then identifies the discrete optical sign orbit, while mechanical regularity determines which members of that orbit can belong to the same physical field class. A likelihood attaches a statistical metric to the same Jacobian, exposing both nuisance-induced information loss and an exact amplitude–sensitivity scale degeneracy. The numerical studies then test these predictions under branch-aware maximum likelihood, including multibranch and nonregular stationary regimes, before an independent phase-referenced vibration channel is used to resolve the remaining sign and scale directions. The resulting operator is also directly usable as a differentiable likelihood layer in physics-consistent inversion.

1.1 Research gap

Time-averaged interferometry has a mature literature on Bessel-fringe formation, beginning with the classical time-averaged vibration response [1, 2], and on regional and variational amplitude recovery, fringe-order tracking, surface-optimization inversion, and Bessel-sign reconstruction [3, 4, 7, 8, 5, 11]. Recent full-field interferometric vibrometry further demonstrates the breadth of frequency-domain deformation recovery from scattering surfaces [6]. Estimation limits are likewise established for interference signals and speckle-based displacement measurements [12, 13, 14], and singular-information and boundary-constrained likelihood asymptotics are classical statistical theory [19, 20, 21]. The present work develops an operator-level account of how the even, nonmonotone J_0^2 inverse fiber interacts with mechanical field regularity, likelihood-dependent Fisher geometry, and calibration

identifiability. Optical symmetry generates sensor-equivalent fields; the mechanical class determines their admissibility; the likelihood weights the surviving local Jacobian; and an independent measurement supplies information along exact sign or scale symmetries. This separation makes the boundary between global structural ambiguity and local statistical conditioning explicit.

1.2 Computational role of differentiability

Differentiable optical forward models support physics-consistent reconstruction in holography and fringe analysis [16, 17, 18]. Here differentiability is used as an enabling property of the calibrated measurement model: a mechanics parameterization W_θ is followed by $\mu(W_\theta)$ and the likelihood $-\log p(I \mid \mu(W_\theta), \eta)$. The analysis below characterizes the structural and statistical geometry inherited by any gradient-based estimator. The numerical evidence uses direct maximum likelihood, providing a classical reference before learned or unrolled estimators are considered as subsequent extensions.

The excitation frequency f is treated as a controlled physical coordinate. At each frequency, W_f defines a separate measurement problem; a frequency sequence can additionally support branch tracking, modal continuity tests, and repeated-measurement diagnostics.

1.3 Contributions

The contribution is organized as one inverse-geometric chain, with a clear hierarchy between established machinery and operator-specific results. The strongest structural result is Proposition 5: for the even time-averaged observable, regular mechanical continuity couples otherwise independent nodal-domain sign assignments and contracts a connected regionwise sign orbit to the global pair $\{W, -W\}$. The remaining steps specialize or combine established tools for this squared-Bessel operator: the exact derivative and curvature locate regular and stationary branches; the standard Fisher pullback gives the operator-specific information density and its zeros; classical nonregular likelihood asymptotics are instantiated at Bessel extrema; and the dependence on ca_i records the elementary optical scale orbit. Branch-aware maximum-likelihood studies test single- and multibranch recovery and the full-field covariance prediction, while a synthetic phase-referenced channel organizes residual sign, phase, and scale information through the scalar Q. Differentiability supplies the computational interface; no novelty is claimed for Fisher/CRLB machinery, singular-information asymptotics, or auxiliary-channel fusion as general methods.

2 Observation levels and modeling assumptions

The statistical model depends on the optical quantity assigned to the random variable I. Four levels are distinguished throughout the paper.

Raw specklegram. The camera records a coherent speckle texture with dark-current and read noise, quantization, finite dynamic range, possible saturation, and illumination nonuniformity.

Time-averaged complex modulation. For harmonic motion, temporal averaging of the phase-modulated interference term produces a Bessel factor $J_0(cW)$.

Second-order correlation/contrast observable. The operator analyzed here is a correlation- or contrast-power statistic proportional to the squared magnitude of that averaged modulation, giving a deterministic mean $A + BJ_0^2(cW)$. The unbiased sample variance of independent time-averaged speckle looks developed in Sec. 6 is one constructive statistic with exactly this mean. Other processed observables enter the same framework after their mean map and fluctuation law have been calibrated. This level is the object of the analytical and synthetic results below.

Statistical measurement model. The final layer assigns a likelihood $p(I \mid \mu(W), \eta)$ to the measured second-order observable. The diagonal Gamma model used in Sec. 5 is an analytical reference for multiplicative speckle fluctuations. A processed ESPI image can follow a different effective likelihood after filtering, normalization, clipping, or spatial averaging; that likelihood is an experimental calibration quantity.

2.1 Admissible fields, optical geometry, and units

Let $\Omega \subset \mathbb{R}^2$ be a bounded specimen surface of finite measure. We write $W : \Omega \to \mathbb{R}$ for a signed real modal-amplitude field in a common temporal phase convention. The analytical domain is

$$\mathcal{X} = L^\infty(\Omega) \cap L^2(\Omega), \qquad \|W\|_{\mathcal{X}} = \|W\|_\infty + \|W\|_2. \quad (2)$$

This choice controls pointwise optical phase and field energy and includes the finite modal parameterizations used below.

For an out-of-plane displacement observed along the surface normal and illuminated at a local angle $\theta(x)$ from that normal, the phase sensitivity magnitude is

$$c(x) = \frac{2\pi}{\lambda}\,[1 + \cos\theta(x)]. \quad (3)$$

Thus $[c] = \text{length}^{-1}$, $[W] = \text{length}$, and cW is a dimensionless optical phase. More general illumination-observation geometries replace Eq. (3) by the projection of the optical sensitivity vector onto the displacement direction; the scalar analysis is unchanged after that projection has been calibrated.

The baseline coefficients satisfy

$$A, B, c \in L^\infty(\Omega), \quad A(x) \geq A_{\min} > 0, \quad (4)$$

$$0 \leq B(x) \leq B_{\max}, \qquad 0 < c(x) \leq c_{\max}. \quad (5)$$

The quantities A, B, and μ share the units of the selected correlation/contrast statistic. These conditions keep the mean response positive and the Fréchet derivative bounded.

2.2 From harmonic phase modulation to the squared-Bessel observable

Let the optical phase modulation induced by a harmonic displacement be

$$\Delta\phi(x,t) = \Delta\phi_0(x) + c(x)W(x)\cos(2\pi f t + \psi). \quad (6)$$

The period-averaged complex modulation factor is

$$\begin{aligned}\gamma_{\mathrm{TA}}(x) &= \frac{1}{T}\int_0^T \exp[ic(x)W(x)\cos(2\pi f t + \psi)]\,dt \\ &= J_0(c(x)W(x)),\end{aligned} \quad (7)$$

using the standard integral representation of J_0. The second-order correlation/contrast observable considered in this work is affine in $|\gamma_{\mathrm{TA}}|^2$, hence

$$\mu(x) = A(x) + B(x)|\gamma_{\mathrm{TA}}(x)|^2 = A(x) + B(x)J_0^2(c(x)W(x)). \quad (8)$$

This derivation fixes the observation level to which the subsequent Gamma likelihood applies. Raw time-averaged camera intensity and alternative ESPI subtraction/addition schemes have their own deterministic response functions and can be incorporated by replacing Eq. (8) before the same estimation-theoretic analysis is applied [2, 9].

Dark extrema of the selected statistic occur at zeros of J_0, whereas $W = 0$ is a bright stationary point because $J_0(0) = 1$.

Definition 1 (mean ESPI Bessel operator, MESPI). The mean measurement operator is the Nemytskii map

$$\mathcal{M} : \mathcal{X} \to L^2(\Omega), \qquad \mathcal{M}(W) = A + BJ_0^2(cW). \tag{9}$$

We abbreviate this operator, and the optical channel it describes, as MESPI. The abbreviation is used below to distinguish quantities carried by the time-averaged ESPI observation alone from those supplied by the auxiliary channel of Sec. 7.

3 Differentiability, curvature, and local branch structure

Proposition 1 (Fréchet differentiability on admissible fields). Under the assumptions of Sec. 2, the operator

$$\mathcal{M} : \mathcal{X} \to L^2(\Omega), \qquad \mathcal{M}(W) = A + BJ_0^2(cW), \tag{10}$$

is Fréchet differentiable. Its derivative at W is the bounded multiplication operator

$$[D\mathcal{M}(W)h](x) = g_W(x)h(x), \tag{11}$$

with

$$g_W(x) = -2c(x)B(x)J_0(c(x)W(x))J_1(c(x)W(x)). \tag{12}$$

On bounded subsets of $\mathcal{X}$, the derivative depends continuously on W.

Proof. For fixed x, define $q_x(s) = A(x)+B(x)J_0^2(c(x)s)$. The scalar map is smooth and its first two derivatives are uniformly bounded on bounded subsets of s. Taylor expansion gives

$$q_x(W + h) - q_x(W) - q_x'(W)h = R_x(W, h), \tag{13}$$

with $|R_x(W, h)| \le C|h|^2$. Since $h \in L^\infty(\Omega) \cap L^2(\Omega)$,

$$\|R(W, h)\|_2 \le C\|h\|_\infty\|h\|_2 = o(\|h\|_{\mathcal{X}}). \tag{14}$$

Equation (12) follows from $J_0'(z) = -J_1(z)$. □

The first-order sensitivity vanishes when

$$J_0(cW)J_1(cW) = 0. \tag{15}$$

Zeros of J_0 are dark minima of the squared-Bessel observable; nonzero zeros of J_1 are bright maxima, and $W = 0$ is the central bright stationary point. Gradient-based inversion is therefore locally ill-conditioned near these sets even after a Bessel branch has been selected. Figure 1(a) shows the mean response, the optical gain, and the Gamma information density in the shared optical-phase coordinate: all three vanish on the same stationary set, and the intervening monotonic intervals are the magnitude branches of Proposition 3. Figure 1(b) makes the resulting pointwise multiplicity explicit and is discussed as Study D in Sec. 8.3.

Proposition 2 (stationary-point curvature). Let $z = cW$. The pointwise second derivative of the mean response is

$$\frac{\partial^2\mu}{\partial W^2} = c^2B\left[2J_1^2(z) - J_0^2(z) + J_0(z)J_2(z)\right]. \tag{16}$$

At a location with $B > 0$, a dark extremum $J_0(z_\star) = 0$ satisfies

$$\mu''(W_\star) = 2c^2BJ_1^2(z_\star) > 0, \tag{17}$$

whereas at a nonzero bright extremum $J_1(z_\star) = 0$,

$$\mu''(W_\star) = -2c^2BJ_0^2(z_\star) < 0. \tag{18}$$

At the nodal point $W = 0$, $\mu''(0) = -Bc^2$. Hence a first-order information zero generally retains a quadratic optical response. Standard regular first-order Fisher approximations become nonuniform in the immediate neighborhood of these stationary points.

Proof. Differentiating Eq. (12) gives $-2c^2B[J_0'(z)J_1(z) + J_0(z)J_1'(z)]$. Using $J_0' = -J_1$ and $J_1'(z) = [J_0(z)-J_2(z)]/2$ yields Eq. (16), including the positive J_0J_2 term. At a nonzero zero of J_1, the recurrence $J_0(z) + J_2(z) = 2J_1(z)/z$ gives $J_2 = -J_0$, hence $\mu'' = -2c^2BJ_0^2$. The value at $z = 0$ follows by continuity or from $J_0(z) = 1 - z^2/4 + O(z^4)$. □

Proposition 3 (local branch invertibility). Let $q(z) = J_0^2(z)$ for $z \ge 0$, and let $\{\zeta_m\}$ be the ordered stationary points generated by the zeros of J_0J_1, including $\zeta_0 = 0$. On every open interval (ζ_m, ζ_{m+1}), $q'(z)$ has fixed sign and the map is one-to-one. A pointwise magnitude $|W|$ is therefore locally differentiably recoverable after selecting one such Bessel branch and requiring $g_W \neq 0$. A single intensity value can intersect several branches, producing additional global alternatives even after sign has been factored out.

Corollary 3.1 (finite magnitude-branch fiber on a bounded phase range). For $r \in [0, 1]$ and finite $Z > 0$, define

$$\mathcal{B}_r(Z) = \{z \in [0, Z] : J_0^2(z) = r\}. \tag{19}$$

Then $\mathcal{B}_r(Z)$ is finite. On each open monotonic interval $(\zeta_m, \zeta_{m+1}) \cap [0, Z]$ it contains at most one point; additional solutions can occur only at the finitely many stationary endpoints. Thus, over any bounded physically admissible optical-phase range, magnitude-branch multiplicity is a finite discrete component of the pointwise inverse fiber, separate from the sign-equivalence set.

Proof. The zeros of J_0 and J_1 have no finite accumulation point, so only finitely many stationary points ζ_m lie in $[0, Z]$. Proposition 3 makes J_0^2 one-to-one on each intervening open interval, giving at most one solution there; adding the finite set of endpoints preserves finiteness. □

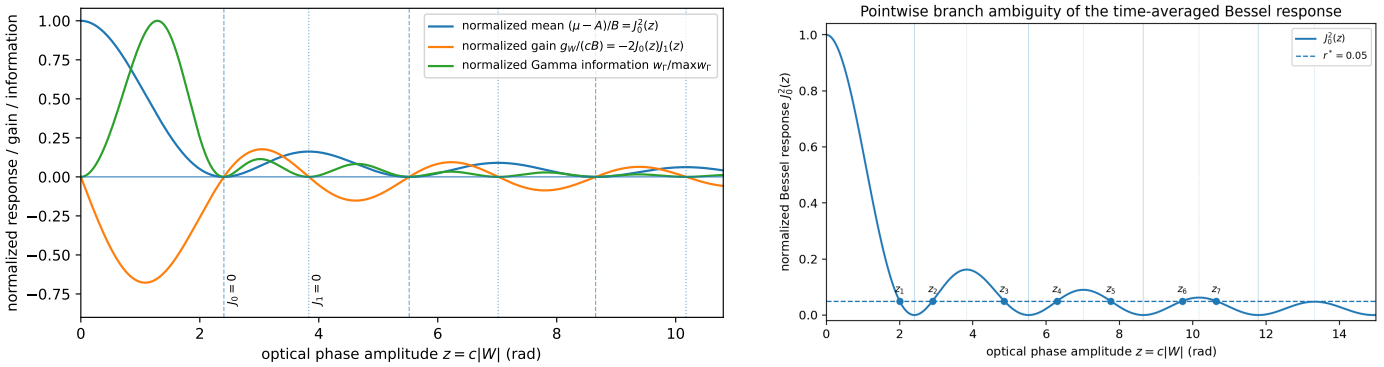
The quarter-power rate is a classical consequence of non-regular likelihood geometry when Fisher information vanishes and the first informative local term is of higher order [19, 20]. Boundary projection and non-Gaussian limiting behavior are likewise standard in constrained maximum likelihood [21]. The following proposition records the squared-Bessel selected-branch specialization used in Study F3.

Proposition 4 (stationary branch rate for the squared-Bessel operator). Let $W_\star$ be a stationary point of a selected adjacent branch with $\mu'(W_\star) = 0$ and $\mu''(W_\star) \neq 0$. If a mean perturbation satisfies $\delta\mu_N = O_p(N^{-1/2})$ and W_N is an adjacent-branch solution of $\mu(W_N) - \mu(W_\star) = \delta\mu_N + o_p(N^{-1/2})$, then

$$|W_N - W_\star| = O_p(N^{-1/4}). \tag{20}$$

For branch-constrained estimators, perturbations outside the admissible response range are projected to the branch boundary, producing a nonregular boundary component in addition to the quarter-power interior scale.

Proof. Taylor expansion at $W_\star$ gives $\mu(W_\star + \delta) - \mu(W_\star) = \frac{1}{2}\mu''(W_\star)\delta^2 + o(\delta^2)$. Balancing this quadratic response with an $O_p(N^{-1/2})$ mean perturbation gives $\delta^2 = O_p(N^{-1/2})$ and therefore $|\delta| = O_p(N^{-1/4})$. □



(a) (b)

Figure 1: Operator geometry in optical-phase coordinates $z = c|W|$. (a) The normalized mean, optical gain, and Gamma information density share the same stationary structure: zeros of J_0 and J_1 delimit monotonic response branches and are exact first-order information zeros. The Gamma curve uses $A = 1$, $B = 0.8$ and is normalized by its maximum; the locations of the zeros are independent of that statistical normalization. (b) Study D: a single normalized response $J_0^2(z) = 0.05$ intersects 7 amplitude values over the plotted range, the pointwise magnitude-branch fiber of Corollary 3.1.

4 Structural sign ambiguity and the full inverse fiber

Let the nodal set of a continuous or piecewise-smooth field be $Z(W) = \{x : W(x) = 0\}$, and suppose that $\Omega \setminus Z(W)$ has finitely many connected components $R_1, \ldots, R_{N_{\text{nodal}}}$. Finiteness holds for the analytic and finite-modal fields considered here and is assumed throughout. The full regionwise sign group is

$$G(W) = \{\pm 1\}^{N_{\text{nodal}}}. \tag{21}$$

For $s = (s_1, \ldots, s_{N_{\text{nodal}}}) \in G(W)$, the action is $[sW](x) = s_m W(x)$ on R_m and $[sW](x) = 0$ on $Z(W)$. Mechanical regularity, boundary conditions, or a finite modal parameterization can exclude some actions of this group from the selected physical field class $\mathcal{X}_{\text{phys}} \subseteq \mathcal{X}$. We therefore define the physically admissible subset

$$G_{\text{adm}}(W) = \{s \in G(W) : sW \in \mathcal{X}_{\text{phys}}\}. \tag{22}$$

The set $G_{\text{adm}}(W)$ need not be a subgroup for an arbitrary mechanical class; the optical equivalence result requires only this admissible subset.

Theorem 1 (admissible sign-equivalence non-identifiability). The mean ESPI Bessel operator factors through pointwise magnitude,

$$\mathcal{M}(W) = A + BJ_0^2(c|W|). \tag{23}$$

Consequently,

$$\mathcal{M}(sW) = \mathcal{M}(W) \qquad \forall s \in G_{\text{adm}}(W). \tag{24}$$

Define the admissible sensor-equivalent sign set

$$\mathcal{E}_{\text{sign}}(W) = \{sW : s \in G_{\text{adm}}(W)\}. \tag{25}$$

Because W is nonzero on each connected component R_m, distinct regionwise sign assignments produce distinct fields, and therefore

$$|\mathcal{E}_{\text{sign}}(W)| = |G_{\text{adm}}(W)| \le 2^{N_{\text{nodal}}}. \tag{26}$$

Equality holds when every independent nodal-region sign reversal remains in $\mathcal{X}_{\text{phys}}$. The optical sensor is insensitive to every member of $\mathcal{E}_{\text{sign}}(W)$; the mechanical model determines how much of the combinatorial upper bound is physically realizable.

Proof. Evenness of J_0^2 gives $J_0^2(csW) = J_0^2(cW)$ pointwise for every $s \in G_{\text{adm}}(W)$. The full group $G(W)$ contains exactly two sign choices on each connected nodal region, so $|G(W)| = 2^{N_{\text{nodal}}}$. Restriction to the admissible subset gives the upper bound in Eq. (26). If every group action remains mechanically admissible, $G_{\text{adm}}(W) = G(W)$ and equality follows. □

Proposition 5 (regular-interface contraction of the sign orbit). Build a graph whose vertices are the nodal domains R_m. Join two vertices when their closures share an open C^1 nodal arc Γ on which the normal derivative of W is nonzero. Let this regular-interface graph have C connected components. If $W \in C^1(\Omega)$ and a regionwise signed field sW also belongs to $C^1(\Omega)$, then the assigned sign is constant on each graph component. Hence

$$|\mathcal{E}_{\text{sign}}^{C^1}(W)| \le 2^C. \tag{27}$$

When the graph is connected, the mechanically regular sensor-equivalent set is exactly $\{W, -W\}$.

Proof. On a regular nodal interface $W = 0$ while the one-sided normal derivatives are nonzero. Multiplication by signs s_m and s_n changes the normal derivatives on the two neighboring domains by those same factors. Continuity of $\nabla(sW)$ across the interface requires $s_m = s_n$. Sign equality propagates along every path of the graph, leaving one independent sign per connected component. □

For a bending-energy field class, a non-global flip across a regular nodal arc has an even stronger consequence. The first derivative jumps across the interface, so the distributional second derivative contains an interface term. Such a field is outside H^2 and has no finite continuum bending energy. This observation is used in Study C1 below. For fourth-order plate fields, regular-interface adjacency is retained as an explicit hypothesis; the separable plate field used there satisfies it analytically.

Remark (full inverse fiber). Define

$$\mathcal{F}(W) = \{\widetilde{W} : \mathcal{M}(\widetilde{W}) = \mathcal{M}(W)\}. \tag{28}$$

Then

$$\mathcal{E}_{\text{sign}}(W) \subseteq \mathcal{F}(W). \tag{29}$$

The inclusion can be strict because different magnitudes on different Bessel branches satisfy

$$J_0^2(c|W_1|) = J_0^2(c|W_2|), \qquad |W_1| \ne |W_2|. \tag{30}$$

The admissible sign-equivalence set and branch alternatives are therefore separate components of global non-identifiability. Fringe-order information, controlled amplitude sweeps, multi-frequency continuation, independent amplitude references, and mechanical priors can contract this inverse fiber.

Local sensitivity has a different mathematical status. Around a selected branch, the linearized map is $h \mapsto g_W h$. In a finite parameterization, local metric identifiability is governed by the smallest eigenvalue of the Fisher matrix. Full rank establishes local observability, while the magnitude of that smallest eigenvalue determines stability. This distinction is used in the numerical conditioning and inversion studies below.

5 Likelihood geometry and Fisher information

The optical operator fixes the Jacobian through which mechanical parameters enter an observation. The likelihood determines the statistical metric placed on that Jacobian. Let a calibrated scalar observation at x have a regular density $p(I \mid \mu, \eta)$, with nuisance descriptor $\eta(x)$ fixed with respect to the mechanical parameter vector θ. Define the Fisher information of the observation family with respect to its mean coordinate by

$$\mathcal{I}_\mu(\mu, \eta) = \mathbb{E}\big[(\partial_\mu \log p(I \mid \mu, \eta))^2\big]. \tag{31}$$

5.1 Likelihood-general conditional Fisher matrix

The next identity is the standard Fisher pullback through a differentiable mean map; it is stated to expose the squared-Bessel operator factor separately from the likelihood weighting [12, 13, 14].

Fisher pullback identity. Assume conditionally independent observations with sampling density $\rho(x)$, normalized so that $\int_\Omega \rho\, dx = P$ equals the number of sampled measurements, and suppose the mechanical parameters enter the observation model through the calibrated mean field $\mu_\theta = A + BJ_0^2(cW_\theta)$. Then

$$F_{ij}(\theta) = \int_\Omega \rho(x)\, \mathcal{I}_\mu[\mu_\theta(x), \eta(x)]\, g_W(x)^2 \frac{\partial W}{\partial \theta_i} \frac{\partial W}{\partial \theta_j}\, dx. \tag{32}$$

The information density therefore separates into a likelihood-dependent scalar metric and the operator/parameterization factor $g_W^2 \partial_i W \partial_j W$.

Proof. By the chain rule, $\partial_{\theta_i} \log p = (\partial_\mu \log p)\partial_{\theta_i}\mu$ and $\partial_{\theta_i}\mu = g_W \partial_{\theta_i} W$. Taking the expected score product and integrating over the sampling density gives Eq. (32). □

Sign symmetry, Bessel branch multiplicity, and the stationary sets of the mean operator are independent of the likelihood. Statistical assumptions change the positive weighting $\mathcal{I}_\mu$ applied away from those sets. The same separation also makes the scale-confounding result below likelihood-general whenever the mechanical state enters only through the mean field.

5.2 Gamma mean–shape specialization

A Gamma mean–shape model gives a closed-form reference for multiplicative fluctuations of a calibrated positive observable [10]:

$$I(x) \sim \Gamma\left(k(x), \frac{\mu(x)}{k(x)}\right), \tag{33}$$

with

$$\mathbb{E}[I] = \mu, \qquad \mathrm{Var}(I) = \frac{\mu^2}{k}. \tag{34}$$

Here the second Gamma argument is the scale parameter and $k(x)$ is an effective number of independent samples at the measurement scale.

Gamma mean-coordinate information. For $I \sim \Gamma(k, \mu/k)$ with fixed k,

$$\mathcal{I}_\mu = \frac{k}{\mu^2}. \tag{35}$$

Proof. Up to terms independent of μ, $\ell(\mu) = -kI/\mu - k\log\mu$, and therefore $-\mathbb{E}[\partial_\mu^2 \ell] = k/\mu^2$. □

Gamma mean–shape orthogonality. If both μ and k are parameters of the same Gamma mean–shape family,

$$F_{\mu k} = 0, \qquad F_{kk} = \psi_1(k) - \frac{1}{k}, \tag{36}$$

where ψ_1 is the trigamma function. Coupling to k appears when k depends on the mechanical parameters or when a different correlated or mixed-noise likelihood is used.

For a modal field $W_\theta = \sum_i a_i \varphi_i$, Eq. (32) becomes

$$F_{ij}^\Gamma = \int_\Omega \rho(x) \frac{k(x) g_W(x)^2}{\mu(x)^2}\, \varphi_i(x)\varphi_j(x)\, dx, \tag{37}$$

and the local information density per unit sampled measure is

$$w_\Gamma(x) = \frac{k(x) g_W(x)^2}{\mu(x)^2} = \frac{4c^2 B^2 k J_0^2(cW) J_1^2(cW)}{[A + BJ_0^2(cW)]^2}. \tag{38}$$

The factor $1/\mu^2$ redistributes information toward lower-response neighborhoods; the exact information zeros remain fixed by $g_W = 0$.

Discrete computational form. For P sampled pixels and a modal design matrix $\Phi \in \mathbb{R}^{P\times p}$ with $W = \Phi \boldsymbol{a}$, the field Jacobian is diagonal in the pixel basis and

$$J_{\boldsymbol{a}} = \mathrm{diag}(\boldsymbol{g})\Phi, \qquad F_{\boldsymbol{a}} = \Phi^T \mathrm{diag}\left(\rho_p \frac{k_p g_p^2}{\mu_p^2}\right)\Phi. \tag{39}$$

Thus Jacobian-vector and adjoint-Jacobian products can be evaluated without forming a dense image-to-image Jacobian; the computational structure is pointwise, while the principal inverse difficulty is branch and parameter identifiability. For a regular unbiased modal estimator with nonsingular $F_{\boldsymbol{a}}$,

$$\mathrm{Cov}(\widehat{\boldsymbol{a}}) \succeq F_{\boldsymbol{a}}^{-1}, \qquad \mathrm{Var}(\widehat{W}_p) \geq \boldsymbol{\phi}_p^T F_{\boldsymbol{a}}^{-1} \boldsymbol{\phi}_p, \tag{40}$$

where $\boldsymbol{\phi}_p^T$ is row p of Φ. Equation (40) therefore predicts a spatial uncertainty field from the global modal Fisher geometry rather than from the scalar weight $w_\Gamma(p)$ alone. Study F4 tests this prediction directly.

5.3 Nuisance-aware Fisher matrix and calibration identifiability

For a finite-dimensional nuisance vector ν that enters the mean response, write $\vartheta = (\theta, \nu)$. Under the Gamma specialization,

$$F_{\alpha\beta}(\vartheta) = \int_\Omega \rho(x) \frac{k(x)}{\mu(x)^2} \frac{\partial \mu}{\partial \vartheta_\alpha} \frac{\partial \mu}{\partial \vartheta_\beta}\, dx. \tag{41}$$

For global scalar calibration parameters,

$$\frac{\partial \mu}{\partial A} = 1, \qquad \frac{\partial \mu}{\partial B} = J_0^2(cW), \qquad \frac{\partial \mu}{\partial c} = \frac{W}{c} g_W. \tag{42}$$

If $F_{\nu\nu}$ is nonsingular, the information retained for θ after profiling ν is the Schur complement

$$F_{\theta|\nu} = F_{\theta\theta} - F_{\theta\nu} F_{\nu\nu}^{-1} F_{\nu\theta}. \tag{43}$$

Observation (exact scale orbit of c and linear modal amplitudes). For $W = \sum_i a_i \varphi_i$ the mean response depends on the sensitivity and the modal amplitudes only through the products ca_i. The reparameterization $(c, \boldsymbol{a}) \mapsto (qc, \boldsymbol{a}/q)$ therefore leaves μ pointwise invariant for every $q > 0$, which is immediate from Eq. (8). Equivalently, in derivative form,

$$c\frac{\partial \mu}{\partial c} = \sum_i a_i \frac{\partial \mu}{\partial a_i}, \tag{44}$$

so the Jacobian columns are linearly dependent and every Fisher matrix formed as $J^T D J$ with positive mean-coordinate weighting is singular along

$$\boldsymbol{v}_0 = (a_1, \ldots, a_p, -c)^T. \tag{45}$$

The Bessel observation identifies optical-phase products, not metric amplitudes.

We record this elementary invariance not as a result in itself but because it fixes exactly what an auxiliary measurement must supply. The Cramér–Rao bound is infinite along $\boldsymbol{v}_0$ and finite in its orthogonal complement, so no amount of optical data resolves the scale, while any independent channel with a nonzero component along $\boldsymbol{v}_0$ resolves it completely. Section 7 makes that requirement quantitative: the curvature restored along $\boldsymbol{v}_0$ is a single measurable design number, and the same number controls global sign discrimination.

In the three-mode configuration used below, profiling unknown global A and B retains 48.5% of the conditional Fisher trace and 3.72% of its smallest positive eigenvalue. Adding c as an unknown produces the predicted numerical null, with normalized residual 2.31×10^{-16}. These values describe that configuration; the exact scale identity itself is structural.

5.4 Interpretation of the local information density

For additive Gaussian noise with fixed variance σ^2, $\mathcal{I}_\mu = 1/\sigma^2$; under the Gamma reference, $\mathcal{I}_\mu = k/\mu^2$. The two models therefore use the same optical gain and differ in how response levels are weighted. This distinction is useful for experimental design, but a global optimum additionally depends on the parameter basis, excitation family, spatial sampling, and design criterion. The numerical studies below focus on quantities that survive those choices: the location of information zeros, the exact scale null, and the behavior of estimators on regular and nonregular branches.

6 Extensions beyond the diagonal Gamma reference

The diagonal Gamma likelihood is a calibration model for the positive second-order observable in Sec. 2. Camera response, read noise, clipping, quantization, finite-speckle correlation, and spatial filtering can change the observation family. A correlated Gaussian likelihood or local Gaussian quasi-likelihood, for example, contributes

$$F_{ij}^{\mathrm{corr}} = \left(\frac{\partial \boldsymbol{\mu}}{\partial \theta_i}\right)^T \Sigma^{-1} \left(\frac{\partial \boldsymbol{\mu}}{\partial \theta_j}\right), \tag{46}$$

with covariance-derivative terms when Σ depends on θ. A correlated Gamma field requires a joint probability model rather than a covariance matrix alone. In all of these cases, the deterministic Jacobian g_W remains the optical factor through which W enters the mean.

6.1 Microscopic-speckle robustness of the Gamma weighting

An explicit time-averaged speckle construction provides a second statistical view of the same deterministic operator. One independent look is

$$Y = X + I_r + 2\sqrt{X I_r}\, J_0(z)\cos\phi, \qquad z = c|W|, \tag{47}$$

with $X \sim \mathrm{Exp}(\bar{I}_o)$, $\phi \sim \mathrm{Unif}(0, 2\pi)$, and independent X and ϕ. Take the unbiased sample variance S_M of M independent looks as the second-order observable. Its mean is exactly

$$\mathbb{E}[S_M] = \bar{I}_o^2 + 2 I_r \bar{I}_o J_0^2(z) = A + B J_0^2(z), \tag{48}$$

where $A = \bar{I}_o^2$ and $B = 2 I_r \bar{I}_o$. The fourth central moment of one look can be written in terms of $\mu = \mathbb{E}[S_M]$ as

$$\mu_{4,Y} = 3\mu^2 + 12A\mu - 6A^2, \tag{49}$$

which yields

$$\mathrm{Var}(S_M) = \frac{2\mu^2}{M-1} + \frac{12A\mu}{M} - \frac{6A^2}{M} \equiv h_M(\mu). \tag{50}$$

The quantity

$$k_{\mathrm{var}}(\mu) = \frac{\mu^2}{h_M(\mu)} \tag{51}$$

is a moment-equivalent dispersion index. The corresponding $g_W^2/h_M(\mu)$ is used as a variance-based quasi-information diagnostic. It preserves the same deterministic information zeros while allowing the relative weighting across response levels to differ from fixed-shape Gamma.

For $A = 1$, $B = 0.8$, and $M = 16$, k_{var} changes from 1.9672 at $\mu = A$ to 2.3028 at $\mu = A + B$. The fixed-Gamma low/high response-level weighting ratio is 3.24, whereas the microscopic variance gives 2.7679, a 14.6% reduction. On the first monotonic Bessel branch the maximum of the normalized local weighting shifts only from $z = 1.2862$ to $z = 1.2568$, a shift of -2.29%. The main effect is a modest redistribution of information; the optical stationary sets remain unchanged.

7 Residual sign and scale resolution with phase-referenced information

Mechanical regularity can contract the regionwise optical sign orbit, but a connected regular-interface graph still leaves the pair $\{W, -W\}$. The Bessel channel also retains the amplitude–sensitivity scale orbit identified by the scale-orbit observation of Sec. 5. A phase-referenced vibration measurement addresses both residual directions directly.

7.1 Complex velocity observation

At sample locations x_m, consider the auxiliary complex velocity model

$$y_m = e^{i\psi} i\omega W(x_m) + \epsilon_m, \qquad \epsilon_{m,R}, \epsilon_{m,I} \overset{\mathrm{iid}}{\sim} \mathcal{N}(0, \sigma_v^2). \tag{52}$$

The phase origin ψ is referenced to the excitation. Define

$$S_M = \sum_{m=1}^{M} W(x_m)^2, \qquad Q = \frac{\omega^2 S_M}{\sigma_v^2}. \tag{53}$$

When the external phase origin is fixed modulo 2π, the two global signs have mean vectors $\pm\boldsymbol{u}$, and for equal priors

$$D_{\mathrm{KL}} = 2Q, \qquad d^2 = 4Q, \qquad P_e = \Phi(-\sqrt{Q}). \tag{54}$$

Here d^2 is the squared Mahalanobis separation under the complex Gaussian noise convention in Eq. (52).

For a modal field $W = \sum_i a_i \varphi_i$,

$$\frac{\partial \nu_m}{\partial a_i} = i\omega e^{i\psi} \varphi_i(x_m), \qquad \frac{\partial \nu_m}{\partial c} = 0, \tag{55}$$

where $\nu_m = e^{i\psi} i\omega W(x_m)$. Along the MESPI scale-null vector $\boldsymbol{v}_0$,

$$\boldsymbol{v}_0^T F_V \boldsymbol{v}_0 = Q. \tag{56}$$

Thus any informative probe set with $S_M > 0$ adds curvature along the optical scale orbit.

If ψ is estimated locally,

$$\frac{\partial \nu_m}{\partial \psi} = -\omega e^{i\psi} W(x_m), \tag{57}$$

and the real Fisher metric gives $F_{a_i \psi} = F_{c\psi} = 0$ and $F_{\psi\psi} = Q$. This local orthogonality does not remove the global symmetry

$$(W, \psi) \sim (-W, \psi + \pi). \tag{58}$$

Indeed, for $\boldsymbol{u} = i\omega[W(x_1), \ldots, W(x_M)]^T$,

$$\min_{\psi} \|\boldsymbol{y} - e^{i\psi}\boldsymbol{u}\|^2 = \|\boldsymbol{y}\|^2 + \|\boldsymbol{u}\|^2 - 2|\langle \boldsymbol{u}, \boldsymbol{y}\rangle|, \tag{59}$$

which is identical for $\boldsymbol{u}$ and $-\boldsymbol{u}$ realization by realization. A full-rank local Fisher matrix can therefore coexist with a global sign symmetry when the phase origin is free.

Phase-reference design identity. Along the scale family $a(q) = a_0/q$, $c(q) = qc_0$, the MESPI mean is constant. The expected auxiliary loss increment is

$$\mathbb{E}[\Delta \mathcal{L}_V(q)] = \frac{Q}{2}\left(\frac{1}{q} - 1\right)^2. \tag{60}$$

Consequently,

$$\boxed{\begin{aligned} Q = \frac{D_{\mathrm{KL}}}{2} = \frac{d^2}{4} = \boldsymbol{v}_0^T F_V \boldsymbol{v}_0 = F_{\psi\psi} \\ = \frac{\partial^2}{\partial q^2} \mathbb{E}[\Delta \mathcal{L}_V(q)]\Big|_{q=1}. \end{aligned}} \tag{61}$$

At an exact nodal probe, $S_M = 0$ and every quantity in Eq. (61) vanishes. The auxiliary amplitude block itself can remain nonzero in directions orthogonal to the scale orbit, so a nodal probe is a specific failure of sign/scale resolution rather than an absence of all modal information.

7.2 Continuity across excitation frequencies

A frequency sequence supplies a second source of structure. Under regular dynamics, nodal lines and modal fields evolve continuously with excitation frequency except at genuine modal interactions or abrupt state changes. A sign or branch assignment that is plausible at one frequency can therefore be rejected when it violates cross-frequency continuity. This constraint is complementary to the phase reference: it narrows the admissible inverse fiber but does not by itself create an absolute displacement scale.

7.3 Mechanical priors and finite-element constraints

A Gaussian mechanical prior

$$p(W) \propto \exp\Big[-\frac{\alpha}{2}\|LW\|^2\Big] \tag{62}$$

adds $\alpha L^T L$ to the local posterior precision. Finite-element or sparse vibrometry constraints act in the same way after discretization. Such regularization can select among weakly supported directions when the prior has curvature there. The distinction remains useful: sensor identifiability is determined by the measurement operator, while prior curvature expresses mechanical information supplied from outside that observation.

8 Synthetic diagnostics of the information structure

The analytical results are tested on controlled synthetic fields with known branches, modes, and noise laws. Optical-phase units are used where c is absorbed into the field amplitude; unless stated otherwise the baseline mean is $\mu(W) = 1 + 0.8J_0^2(W)$. The studies are organized by the structural question they test rather than by figure order.

8.1 Study C1: optical sign orbit and mechanical contraction

Consider

$$W(x, y) = 10 \sin(3\pi x) \sin(2\pi y), \qquad (x, y) \in [0, 1]^2. \tag{63}$$

Its zero set forms six nodal domains. Enumerating all $2^6 = 64$ regionwise sign assignments gives exactly the same J_0^2 mean response for every candidate. The same field satisfies

$$-\Delta W = 13\pi^2 W, \qquad \Delta^2 W = (13\pi^2)^2 W, \tag{64}$$

with $W = \Delta W = 0$ on the rectangular boundary. It is therefore a simply-supported/Navier Kirchhoff plate mode for this geometry.

The seven regular internal adjacencies connect all six domains. Proposition 5 then contracts the optical set analytically to

$$64 \longrightarrow 2, \tag{65}$$

with W and $-W$ as the two C^1 representatives. The bending functional

$$R(W) = \int_\Omega (\Delta W)^2 \, dxdy \tag{66}$$

has the exact value 4.1155×10^5. Under centered-grid refinement the smooth field converges to that value, reaching relative error 5.20×10^{-6} on the finest grid. A fixed non-global domain flip grows approximately as $h^{-0.959}$ over the four finest grids, consistent with the derivative jump predicted by the continuum argument. Figure 2 separates the geometric and analytic content of Proposition 5: isolated points of the nodal set at which ∇W vanishes need not be regular, because sign compatibility is transmitted through the open regular arcs connecting adjacent nodal domains.

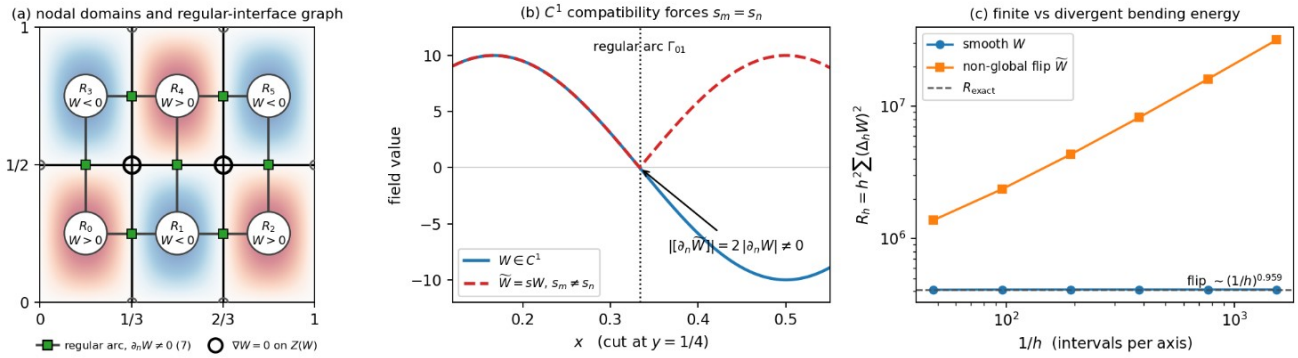


Figure 2: Study C1: regularity-driven contraction of the mechanical sign orbit. (a) The reference field $W = 10\sin(3\pi x)\sin(2\pi y)$ has six nodal domains $R_0, \ldots, R_5$ joined by seven regular open nodal arcs on which $\partial_n W \neq 0$ (squares); the regular-interface graph they define, drawn here on the geometry itself, is connected. Circles mark isolated points of $Z(W)$ at which ∇W vanishes, including the interior nodal crossings and the points where interior nodal lines meet the domain boundary. These isolated points are excluded from the regular-interface hypothesis, which every adjacency satisfies on an open subarc away from them. (b) Local mechanism across one interface. The original C^1 field crosses the arc smoothly, whereas a domain multiplier differing across the interface gives $|[\partial_n \widetilde{W}]| = 2|\partial_n W| \neq 0$, so $\widetilde{W} \notin C^1$. (c) The bending energy of the smooth mode converges to R_{exact}, reaching relative error 5.20×10^{-6} on the finest grid, whereas a fixed non-global flip grows as $(1/h)^{0.959}$, consistent with loss of the finite-bending-energy H^2 class. Connectivity therefore forces a common multiplier $s_0 = \cdots = s_5 = s$, with $s \in \{-1, +1\}$, and contracts the $2^6 = 64$ optical sign assignments to the global pair $\{W, -W\}$. The optical equality of all 64 assignments follows exactly from the even squared-Bessel response and is stated in the text rather than plotted.

8.2 Study C2: statistical weighting under a microscopic speckle model

The microscopic model of Sec. 6.1 preserves the deterministic mean while changing the variance of the second-order statistic. At the baseline $A = 1$, $B = 0.8$, $M = 16$, the moment-equivalent dispersion index moves from 1.9672 at the low-response level to 2.3028 at the high-response level. Fixed-shape Gamma gives a low/high scalar weighting ratio of 3.24; the variance-aware diagnostic gives 2.7679, a 14.6% reduction. The first-branch maximum moves from $z = 1.2862$ to $z = 1.2568$. The direction of the low-response preference survives, but its magnitude belongs to the observation model. Both weightings retain zeros at the same J_0 and J_1 stationary locations. The corresponding weighting curves are shown in the Supplement.

8.3 Study D: explicit Bessel-branch multiplicity

To visualize the part of the inverse fiber that remains after sign is removed, we solve

$$J_0^2(z) = r^*, \qquad z = c|W|, \tag{67}$$

for the normalized response level $r^* = 0.05$ over $0 \le z \le 15$ rad. Seven solutions are present at

$$\begin{aligned} z = {} & 2.0005,\ 2.8981,\ 4.8555,\ 6.2990, \\ & 7.7585,\ 9.7158,\ 10.6382 \text{ rad}. \end{aligned} \tag{68}$$

Figure 1(b) marks these distinct amplitudes on the same squared-Bessel response. The first monotone magnitude branch is $0 < z < j_{0,1} = 2.4048256$; subsequent solutions demonstrate the additional global branch ambiguity characterized in Proposition 3.

8.4 Study E: gradient verification

The implemented derivative is checked independently in three ways over 1001 samples spanning $-5.5 \le W \le 5.5$ rad. The analytical expression in Eq. (12) is compared with automatic differentiation through a 60-term power-series implementation of J_0 and with the central finite difference

$$g_{\text{FD}}(W; h) = \frac{\mu(W + h) - \mu(W - h)}{2h}. \tag{69}$$

The relative ℓ_2 error between the analytic and automatic-differentiation gradients is 5.96×10^{-16}, with maximum absolute error 6.80×10^{-16}. The best finite-difference result occurs at $h = 10^{-5}$, with relative error 4.90×10^{-11}; larger steps exhibit the expected second-order truncation trend and smaller steps enter the roundoff regime. Near exact zeros of g_W, absolute errors are used instead of unstable pointwise relative errors. The full step-size sweep is given in the Supplement.

8.5 Study F: branch-aware differentiable maximum-likelihood inversion

The computational role of the derivative is tested in a three-mode inverse problem. On a 48×48 grid, the field is

$$W_\theta = \sum_{i=1}^{3} a_i \varphi_i, \tag{70}$$

with individually normalized sine modes and

$$\theta_{\text{true}} = (0.58, -0.21, 0.16)^T. \tag{71}$$

The maximum pointwise amplitude is 1.5073 rad, below the first zero $j_{0,1} = 2.4048$ rad. Gamma observations are generated at $k \in \{2, 8, 32\}$ and fitted by L-BFGS-B using the exact likelihood gradient

$$\frac{\partial \mathcal{L}}{\partial a_i} = \sum_p k \frac{\mu_p - I_p}{\mu_p^2}\, g_{W,p} \varphi_i(x_p). \tag{72}$$

Each of 80 Monte Carlo realizations per k is initialized locally around the chosen signed representative. All fits converge, and every recovered field remains inside the first magnitude branch:

Table 1: Representative optimizer comparison for Study F ($k = 8$).

Method	Iter.	Evaluations
L-BFGS-B + analytic gradient	7	10
L-BFGS-B + finite differences	7	40
Powell (derivative-free)	4	232

the largest $\max|W|$ observed across the fitted ensembles is 1.8331 rad.

The aggregate parameter RMSE decreases from 0.0553 to 0.0292 and 0.0138 as k increases. Agreement between empirical dispersion and the local Fisher prediction is quantified in Study F4 below, which repeats this configuration with a 1,000-realization ensemble and reports the comparison at field level; the small-ensemble modal-parameter ratios are retained in the Supplement. The experiment evaluates recovery within a selected Bessel and sign branch; Study D characterizes the global alternatives outside that local basin.

A representative $k = 8$ realization also compares optimization interfaces. The analytic-gradient and finite-difference L-BFGS-B runs converge to the same negative log-likelihood (27430.4468141) and parameter vector to numerical precision. The analytic-gradient run requires 10 objective evaluations, finite-difference L-BFGS-B requires 40, and derivative-free Powell requires 232. Table 1 reports the comparison as function evaluations rather than wall-clock timing so that the result reflects the forward-model workload directly.

Finally, the same three-mode configuration tests the nuisance result in Sec. 5.3. At $k = 8$, profiling unknown global A and B retains 48.5% of the calibrated Fisher trace and 3.72% of the smallest positive eigenvalue. When c is included as an unknown scalar, the joint Fisher spectrum contains a numerical zero and the predicted scale null vector has normalized residual 2.31×10^{-16}. This machine-precision residual is retained only as an implementation sanity check; the structural scale statement follows directly from the invariance in Eq. (44).

8.6 Study F2: multibranch modal recovery

The baseline Study F remains inside the first monotonic interval. To test the larger inverse fiber directly, the same normalized three-mode field is scaled to $\max|W| = 1.5073$, 5, and 8 rad. The number of monotonic intervals of J_0^2 traversed from zero to the maximum field is

$$N_{\mathrm{mono}}(z_{\max}) = 1 + \#\{\zeta \in (0, z_{\max}) : J_0(\zeta)J_1(\zeta) = 0\}, \quad (73)$$

which gives one, three, and five intervals in the three regimes.

A noise-free 64-start Sobol audit returns one sign-aligned optimum cluster in the first-branch and 5-rad cases. At 8 rad, three clusters appear and 50/64 starts reach the true basin. Under Gamma noise with $k = 8$ and 40 paired realizations, local and 16-start multistart optimization recover the sign-aligned field within 10% relative L^2 error in 40/40 realizations at 5 rad and 39/40 at 8 rad. A single broad random start gives 36/40 and 34/40, respectively. The pointwise interval labels are more fragile: their median exact fractions are 0.991 and 0.979. The experiment thus separates modal-field recovery from exact pixelwise branch labeling; the modal constraint can suppress much of the pointwise multiplicity even when several Bessel intervals are traversed. Figure 3 contrasts the two recovery criteria across the three amplitude regimes.

8.7 Study F3: predictive Fisher behavior at regular and stationary points

A scalar selected-branch experiment tests whether the local Fisher curvature predicts estimator dispersion. At regular operating points, 10,000 Monte Carlo realizations with 512 independent Gamma observations per realization give a median empirical-standard-deviation to CRLB ratio of 1.047. This is the regular regime anticipated by Eq. (32).

At exact stationary response levels the first derivative vanishes and the usual first-order expansion changes order. Proposition 4 predicts the $N^{-1/4}$ inversion scale from the nonzero quadratic curvature. For sample sizes from $N = 32$ to $N = 8192$, with 20,000 realizations per cell, the fitted log–log RMSE slopes at the first J_0 and first J_1 stationary points are -0.294 and -0.285 [Fig. 4(c)]. Both are consistent with the quarter-power prediction and clearly separated from the regular $N^{-1/2}$ parameter rate; the residual offset from $-1/4$ is expected, because the selected-branch estimator develops an approximately 50% point mass at the branch boundary and the reported RMSE is therefore a mixture of the interior quarter-power scale and that boundary component.

Figure 4(b) reports the ratio of empirical standard deviation to the local Cramér–Rao bound and falls well below unity at the two stationary points. This is not a violation of the Cramér–Rao inequality. The branch-constrained estimator is biased in those neighborhoods, with roughly half of its mass projected onto the branch boundary, so the unbiased-estimator bound no longer applies. The comparison is retained precisely to locate where the regular bound stops being the relevant object: as Fig. 4(a) shows, the local bound diverges at the stationary points while the realized estimator dispersion remains finite. The vanishing of g_W therefore marks a change of local asymptotic regime, not merely a small Fisher eigenvalue.

8.8 Study F4: predictive full-field Fisher covariance

The scalar F3 experiment tests local regular and nonregular behavior. Study F4 returns to the three-mode first-branch field of Study F and evaluates the field-level covariance prediction induced by the modal Fisher matrix. For $k \in \{2, 8, 32\}$, the truth-level Fisher matrix gives

$$\sigma_{\mathrm{pred}}(x_p) = \sqrt{\boldsymbol{\phi}_p^T F_{\boldsymbol{a}}^{-1} \boldsymbol{\phi}_p}. \quad (74)$$

An independent fixed-seed ensemble of 1,000 Gamma realizations per k is fitted with the same analytic-gradient branch-aware MLE as Study F. The sample covariance of the recovered modal coefficients is then propagated to an empirical field standard-deviation map using the same modal basis.

The median spatial ratios $\sigma_{\mathrm{emp}}/\sigma_{\mathrm{pred}}$ are $(1.069, 1.041, 0.996)$ for $k = (2, 8, 32)$, respectively. Spearman correlations between the predicted and empirical field-SD maps are $(0.998, 0.999, 0.999)$, while median coverage of the nominal 95% field intervals is $(0.937, 0.941, 0.950)$. All 1,000 optimizations converge for each k, and every recovered field remains within the selected first magnitude branch. Figure 5 shows the $k = 8$ spatial maps and pixelwise parity across all three noise levels. The result validates the field covariance induced by the global modal Fisher matrix and provides a predictive field-level test of the information geometry.

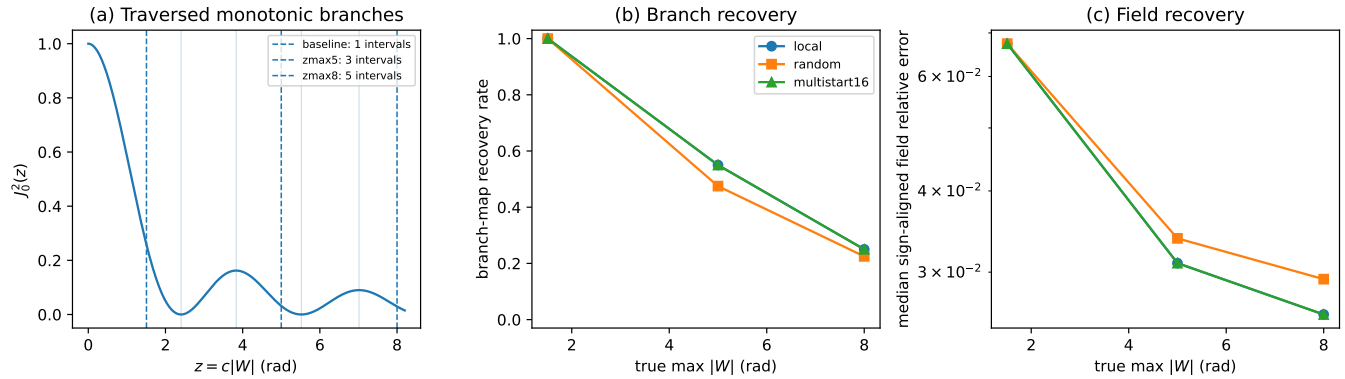


Figure 3: Study F2. (a) The tested amplitude ranges traverse one, three, and five monotonic intervals of J_0^2. (b) Exact branch-map recovery becomes progressively more difficult as stationary boundaries are crossed. (c) The continuous sign-aligned modal field remains accurately recovered in most realizations, especially with local or multistart initialization.

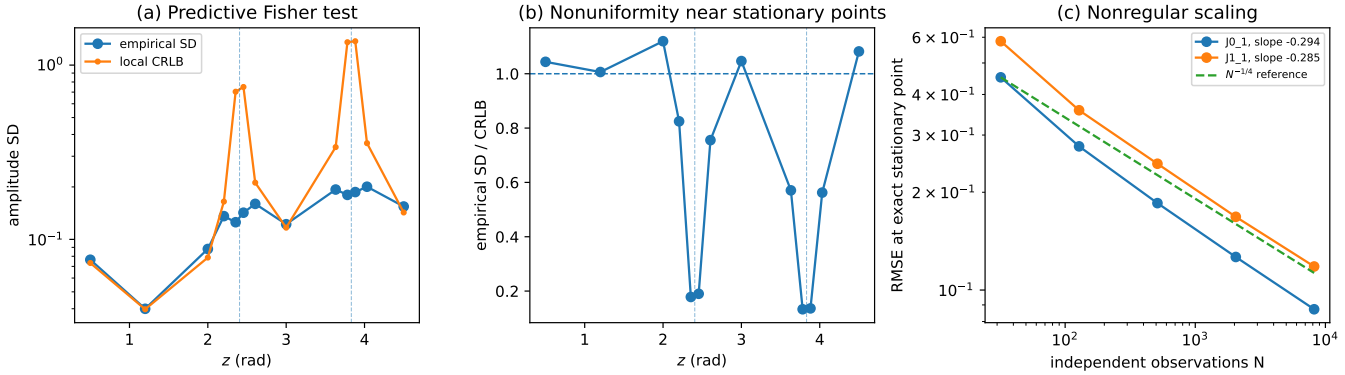


Figure 4: Study F3. (a) Regular selected-branch operating points follow the local Cramér–Rao scale, which diverges at the two stationary points while the realized dispersion stays finite. (b) The empirical-to-CRLB ratio drops below unity there; this reflects the bias and boundary mass of the branch-constrained estimator, not a violation of the bound, which applies only to unbiased estimators. (c) At exact Bessel stationary points RMSE follows an approximately quarter-power law in sample size.

8.9 Study G: phase-referenced resolution of residual sign and scale

Study G applies the phase-referenced model of Sec. 7 to the C1 global sign pair and to the scale-confounded three-mode field of Study F. It has an analytical part (G1–G2), which evaluates the structural identities and ranks in closed form, and an empirical part (G3), which reports oracle binary calibration and operational profile-MLE sign/scale recovery as separate tests.

G1: rank restoration and the exact node. The MESPI Fisher matrix has rank three in (a_1, a_2, a_3, c) and rises to four when the phase-referenced channel is added. If ψ is treated as a local parameter, the rank rises from three to five in (a_1, a_2, a_3, c, ψ), while the global relation $(W, \psi) \sim (-W, \psi + \pi)$ remains exact when the phase origin is free. An analytic cancellation probe at $(x, y) = (0.181258, 0.75)$ has $S_M = 0$; it leaves the joint rank at three even though its auxiliary amplitude block is nonzero. Because $Q \propto S_M = \sum_m W(x_m)^2$ and W vanishes linearly across a regular nodal arc, sign and scale information degrade quadratically in the distance from a nodal line. Exact cancellation is therefore a measure-zero event bounding a smooth practical penalty, and probe placement should be read as a requirement on sampled energy rather than on avoiding isolated points.

This requirement is experimentally actionable without prior knowledge of the signed field once a branch-resolved optical magnitude estimate is available. Such an estimate can be supplied by controlled amplitude continuation, fringe-order anchors, or a mechanical prior. The resulting $|W|$ map identifies high-S_M regions and reconstructed mechanical nodes before the auxiliary probe is placed. The amplitude-aware design below therefore uses only information available from the optical inversion stage, not the signed simulation truth; when the magnitude branch is not yet resolved, probe placement must remain prior- or continuation-guided.

G2: the scale ridge. Following the exact scale orbit $a(q) = a_0/q$, $c(q) = qc_0$, the MESPI mean is flat to 8.88×10^{-16}, whereas the expected joint loss has a unique minimum at $q = 1$ with curvature Q [Fig. 6(c)]. With a fixed full-field RMS noise reference, $\sigma_v = \rho_v \omega W_{\mathrm{RMS}}$, the design variable is governed by the sampled energy S_M rather than by a probe-normalized noise level. The exact relations $D_{\mathrm{KL}} = 2Q$ and $d^2 = 4Q$ are retained symbolically.

G3: empirical sign and scale recovery and the design budget. On the frozen Study-F geometry, 100,000 bal-

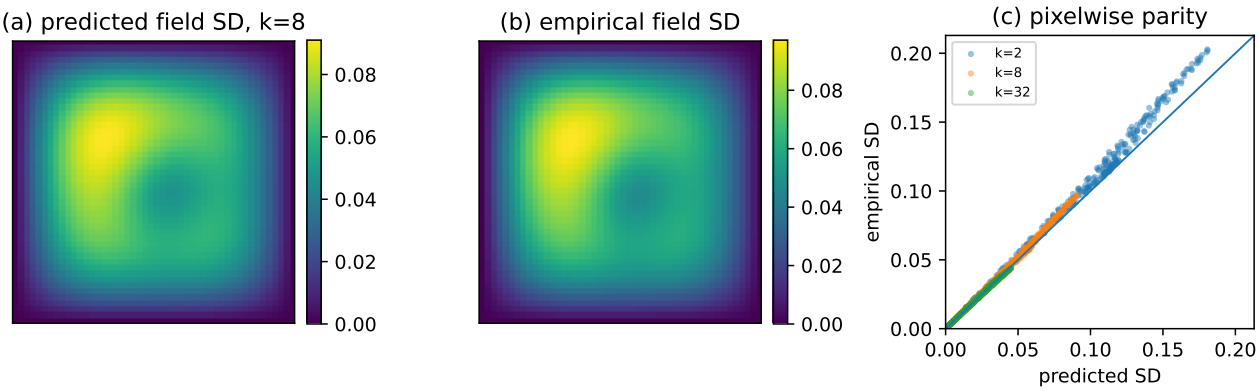


Figure 5: Study F4: predictive full-field covariance. (a) Field standard deviation predicted from $\boldsymbol{\phi}_p^T F_{\boldsymbol{a}}^{-1} \boldsymbol{\phi}_p$ at $k = 8$. (b) Empirical field standard deviation over 1000 independent Gamma-MLE recoveries. (c) Pixelwise parity across $k = 2, 8, 32$. The modal Fisher matrix predicts both the spatial pattern and the scale of regular-regime reconstruction uncertainty.

anced $\pm W$ trials at $Q \in \{1, 2, 3.5, 5.5, 8\}$ calibrate the simple-hypothesis law $P_e = \Phi(-\sqrt{Q})$ to maximum absolute deviation 0.0012, with the theory inside every 99.9% Wilson interval [Fig. 6(a)]. A further 100,000-trial cell at the declared 10^{-3} threshold $Q^* = 9.5495$ gives 0.00114 with 99.9% Wilson interval $[0.000839, 0.001549]$, so the design threshold lies inside the calibrated oracle range.

A separate 1,000-realization operational test first estimates the sign-convention optical-phase vector $\boldsymbol{\beta} = c\boldsymbol{a}$ on the selected first branch, then uses the external channel to select global sign while profiling the positive scale $1/c$. All optical fits converge and remain on that branch. Operational sign error decreases monotonically with Q but does not obey the oracle law exactly. A post-hoc Fisher combination of exact binomial tests gives $X^2 = 35.08$ on 10 degrees of freedom ($p = 1.21 \times 10^{-4}$), with positive excess concentrated at $Q = 2$ (0.03235) and $Q = 3.5$ (0.01532). The same optical fits are reused across Q, so no causal mechanism is assigned to this pattern; the oracle/operational split preceded the authoritative run, whereas the omnibus diagnostic is post-hoc.

Conditional on correct sign, the scale coordinate $\xi = \log(1/c)$ provides a second empirical test. To separate auxiliary noise from the fixed optical-fit ensemble without assuming that interquartile ranges add in quadrature, we use a paired per-realization diagnostic

$$\Delta\xi_Q = \xi_Q - \xi_\infty(\widehat{\boldsymbol{\beta}}), \tag{75}$$

where ξ_∞ is the deterministic zero-auxiliary-noise profile for the same stored optical estimate. Over $Q \in \{20, 50, 200, 1000, 1.7657 \times 10^4\}$, the standard deviation of $\Delta\xi_Q$ follows $Q^{-0.507}$ with log–log $R^2 = 0.9999$, consistent with the regular $Q^{-1/2}$ scale implied by the ridge curvature [Fig. 6(c)]. The optical-only zero-noise ensemble has $\mathrm{SD}(\xi_\infty) = 0.03313$. At the nominal frozen design $Q = 1.7657 \times 10^4$, the paired auxiliary contribution has SD 0.00742 while the total operational SD is 0.03364; the optical floor is therefore 4.47 times larger than the remaining auxiliary component, and the total SD lies only 1.55% above the zero-noise floor. Thus this nominal configuration is already optical-estimation limited with respect to metric-scale precision. The older IQR sweep and its deterministic limit remain in the Supplement as a robust distributional check.

Finally, $Q = \omega^2 S_M/\sigma_v^2$ turns these results into a probe budget. The 10^{-3} oracle target requires only $Q \geq 9.55$, versus $Q = 1.7657 \times 10^4$ for the frozen $M = 3$ design at $\rho_v = 0.03$. Near a regular node $Q(d) = O(d^2)$; at that noise level a single probe crosses the oracle target below 3.92% of the global antinode amplitude [Fig. 6(b)]. This is an oracle exclusion threshold; an end-to-end pipeline can require a larger margin. G3 conditions on successful magnitude-branch localization, while competing-branch and initialization robustness are tested separately in Study F2.

Together the three parts close the synthetic structural chain

$$2^{N_{\mathrm{nodal}}} \longrightarrow 2 \longrightarrow 1, \tag{76}$$

where the first contraction comes from mechanical admissibility and the second from an externally fixed phase reference with $S_M > 0$.

As a robustness check on the MESPI-only non-null geometry, fixed-Gamma and variance-aware weightings were compared under five declared coordinate scalings. The variance-aware matrix is marginally better conditioned in all five: for example $\kappa_+ = 36.5526$ versus 35.5420 under fractional scaling. The effect is small. Under RMS-column scaling each Fisher matrix is separately normalized by $s_j = F_{jj}^{-1/2}$, so that sensitivity case compares two weighting-specific correlation-form coordinate systems.

9 Relation to previous work

The optical kernel belongs to the classical time-averaged interferometry literature. Bessel modulation of vibration fringes and the behavior of several digital speckle observation schemes were established early [1, 2]. Quantitative inversion also has a substantial lineage: Wong and Chan developed full-field vibration-amplitude recovery [3], Borza used regional inversion of the Bessel response [4], Statsenko *et al.* developed phase reconstruction and surface-optimization procedures for time-averaged ESPI [7, 8], and Cywińska *et al.* proposed an end-to-end variational vibration-profilometry pipeline with synthetic and experimental verification [5]. Full-field heterodyne interferometric vibrometry has also extended deformation recovery to broad frequency ranges [6]. The present formulation starts from this measurement and inversion lineage and asks which equivalence classes, metric null directions, and uncertainty fields are induced when the squared-Bessel response is treated as a probabilistic field operator. The numerical evaluation targets operator-specific predictions under a common likelihood—multibranch modal behavior, nonregular stationary scaling, and full-field Fisher

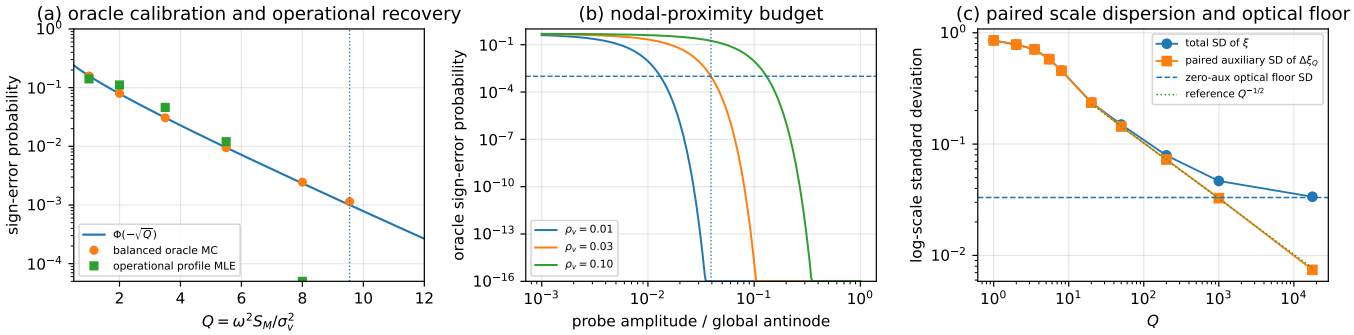


Figure 6: Study G. (a) Balanced oracle binary Monte Carlo, including the declared $P_e = 10^{-3}$ design point, and the separate operational profile-MLE sign-recovery curve. The latter includes optical magnitude estimation and is not assumed to equal the oracle law. (b) Oracle nodal-proximity budget for a single probe, expressed as sampled amplitude relative to the global antinode. (c) Paired metric-scale diagnostic. The total log-scale SD approaches the deterministic zero-auxiliary-noise optical floor, while SD($\Delta\xi_Q$) isolates the auxiliary contribution for the same stored optical estimate; its high-Q rate is $Q^{-0.507}$, close to the $Q^{-1/2}$ reference.

covariance— while established regional and variational methods define the amplitude-recovery context summarized in Table 2.

Existing regional Bessel inversion and Bessel-sign algorithms address fringe order and magnitude-branch selection by tracking monotonic intervals and the sign of the Bessel response [4, 11]. This is distinct from the mechanical signed-field symmetry $W \mapsto -W$ induced by the even squared-Bessel observable. Mathematical nodal-domain theory studies the geometry and counting of sign-constant components of eigenfunctions [22], but it does not by itself characterize sensor-equivalent sign assignments under an even optical map. Proposition 5 concerns that inverse-recovery question: it states when C^1 mechanical continuity couples otherwise independent nodal-domain sign assignments and proves that a connected regular-interface graph leaves exactly the global pair $\{W, -W\}$. To our knowledge, this regularity-driven contraction has not previously been stated in the ESPI inverse-recovery literature; the novelty claimed is the operator/mechanics synthesis, not nodal-domain theory or the existence of sign ambiguity itself.

Phase- or velocity-resolved vibration measurements are also established when an additional temporal reference is supplied. Stroboscopic sideband holography and related heterodyne vibration measurements recover information unavailable to a purely time-averaged even observation [23]. These methods therefore support, rather than contradict, the distinction made here: absolute mechanical direction requires information beyond $J_0^2(cW)$.

The structural analysis then places these operations in an explicit inverse-fiber picture. The regionwise sign group supplies the optical upper set, regular-interface mechanics contracts it, Bessel nonmonotonicity creates a separate magnitude-branch fiber, and a referenced auxiliary channel supplies the final information needed for absolute sign and scale. This separation matters because information supplied at one layer leaves the other layers structurally distinct.

Cramér–Rao bounds and Fisher information are established tools for interference and speckle-displacement measurements [12, 13, 14]. Equation (32) uses that standard pullback machinery for the squared-Bessel ESPI observable; the operator-specific content is the factor g_W^2, its exact stationary zeros, and the resulting modal geometry. The Gamma model supplies the closed-form kg_W^2/μ^2 specialization, while the microscopic-speckle construction checks how much of the spatial weighting survives a different variance law. The nuisance analysis records the elementary scale consequence of the optical argument: sensitivity and linear modal amplitudes enter through the products ca_i.

The quarter-power rate and boundary behavior used in Study F3 belong to the classical theory of singular-information and constrained likelihood problems [19, 20, 21]. The contribution here is to identify the squared-Bessel bright/dark extrema as an optical instance of that nonregular regime and to verify the predicted change of scaling numerically; no new general asymptotic theory is claimed.

Differentiable optical models support physics-consistent reconstruction in holography and fringe analysis [16, 17, 18], while DeepBessel applies learned single-shot vibration-amplitude recovery to time-averaged interference data [15]. The present study does not claim reconstruction-accuracy superiority over regional, variational, or learned amplitude solvers. Their target is amplitude recovery from the measured pattern; the analysis here characterizes sign, branch, stationary, and scale information that is present or absent in the even observation itself. Learned or unrolled estimators are therefore complementary downstream solvers rather than evidentiary baselines for the inverse-geometric claims.

10 Scope and limitations

The analysis concerns a specified positive second-order time-averaged ESPI observable. Applying the same likelihood to a camera experiment requires a calibrated mapping from the recorded speckle data to that observable, including $A(x)$, $B(x)$, $c(x)$ and the effective fluctuation law. Filtering, normalization, finite speckle size, spatial dependence, camera nonlinearity, clipping, read noise, and saturation can change the mean-coordinate Fisher metric even when the deterministic Bessel kernel is preserved.

The Gamma mean–shape likelihood is one calibrated reference. Study C2 shows that an explicit microscopic sample-variance model retains the same mean and information zeros but changes the scalar response-level weighting by a modest amount. Its variance-based quantity is a quasi-information diagnostic rather than an exact likelihood Fisher matrix. Experimental use therefore requires stationarity, dependence, and goodness-of-fit tests at the level of the quantity actually analyzed.

The inversion experiments also distinguish regular local behavior from global and nonregular effects. Study F begins on a chosen sign and first Bessel branch; F2 then introduces additional

Table 2: Relationship to adjacent research lines.

Research line	Established baseline	Analysis developed here
Time-averaged Bessel optics / amplitude inversion	J_0 modulation; regional, variational, and learned amplitude recovery; Bessel-lobe sign and fringe-order selection	Mechanical signed-field inverse fiber under the even J_0^2 observable; Proposition 5 contracts the regular nodal-domain sign orbit to the global pair and separates it from magnitude branches.
Fisher / information operators	Fisher–CRLB pullback, noise-weighted sensitivity Gramians, and local information operators are standard	Squared-Bessel specialization $g_W^2\mathcal{I}_\mu$, exact optical information zeros, nuisance geometry, and a validated full-field covariance prediction.
Nonregular likelihood asymptotics	Singular-information and parameter-boundary rates/distributions are classical	Identification of Bessel bright/dark extrema as the quarter-power optical regime and numerical verification in Study F3.
Phase-resolved auxiliary sensing	Stroboscopic/heterodyne and LDV-type channels supply temporal phase or velocity information	The scalar Q unifies oracle sign separation with phase/scale curvature; operational sign/scale recovery is evaluated separately and remains synthetic.

pointwise branches, F3 shows that first-order Fisher/CRLB asymptotics lose uniform validity at exact stationary responses, and F4 tests the regular modal Fisher prediction at full-field covariance level. The phase-referenced Study G is synthetic. Its G1–G2 results are analytical identities evaluated numerically, whereas G3 is a Monte Carlo test of a realized estimator; the two are reported separately because they carry different evidential weight. An experimental realization must preserve an external phase origin and a traceable polarity convention before its signed output can be interpreted as an absolute displacement direction.

The signed real field W presumes a common temporal phase. For a genuinely complex mode shape with spatially varying phase, the pointwise time average still yields $J_0(c|W(x)|)$, so the mean operator and its Fisher geometry are unchanged; what changes is the sign analysis, because a complex field has no signed real representative and the nodal partition of Sec. 4 is no longer the relevant structure. Lightly damped and well-separated modes satisfy the real convention to good approximation; strongly coupled or heavily damped responses require the phase field itself to be estimated.

The scalar sensitivity c is treated as a global nuisance in the exact scale analysis. Spatially varying calibration fields can introduce additional nuisance directions and should be identified through dedicated calibration measurements. Mechanical priors, cross-frequency tracking, and finite-element constraints can contract the inverse fiber further, but their information is external to the time-averaged optical observation.

11 Experimental calibration and validation requirements

The present evidence is theoretical and synthetic. A quantitative experimental realization separates naturally into optical calibration, likelihood calibration, and inverse validation. The optical stage records the camera operating point, wavelength, illumination/observation geometry, dynamic range, and the transformation from raw specklegrams to the selected second-order statistic. A controlled vibration-amplitude sweep registered to an independent displacement reference can then fit

$$\mu = A + BJ_0^2(cW_{\mathrm{ref}}) \tag{77}$$

across bright, dark, and intermediate response levels. This establishes the measurement scale and the Bessel branch structure before inversion.

Repeated acquisitions at fixed physical conditions calibrate the stochastic model. Their mean–variance relation, spatial dependence, and stationarity show whether a diagonal Gamma effective-sample model is adequate or whether a correlated or mixed-noise likelihood is needed. The same repeated data provide an empirical uncertainty distribution for direct comparison with the local Fisher prediction away from stationary points.

The final stage registers time-averaged ESPI to a phase-referenced SLDV/LDV or an equivalent independent vibration measurement. Absolute sign requires a common phase origin known modulo 2π, documented excitation–response polarity, and a response-direction convention tied to the optical sensitivity vector. A modal phase registration that merely makes a complex field as real as possible is insufficient for this purpose: the transformation $H \mapsto -H$ changes its phase by π and leaves the signed-real eligibility unchanged. These controls make the synthetic Study-G distinction between fixed and free phase experimentally testable.

With those calibrations in place, branch-aware MLE/MAP can be evaluated by modal-parameter error, branch accuracy, nodal agreement, and empirical uncertainty. The exact gradient and Fisher geometry provide a classical reference against which learned or unrolled estimators can then be compared on physical accuracy, uncertainty calibration, and computational cost.

12 Conclusion

The squared-Bessel ESPI observable has three distinct inverse-geometric features. Evenness creates a discrete sign orbit, Bessel nonmonotonicity creates additional magnitude branches, and stationary bright or dark levels remove first-order local sensitivity. The Fréchet derivative and stationary curvature make the third feature explicit, while the nodal-domain analysis separates the first two at the field level.

Mechanical regularity acts before statistical inversion. For a connected regular-interface adjacency graph, a C^1 field can retain only the global pair $\{W, -W\}$ from the regionwise optical orbit. The six-domain plate example therefore contracts 64 sensor-equivalent sign assignments to two exactly; grid refinement confirms that a non-global flip leaves the finite-bending continuum class. The likelihood then supplies a local metric on the same optical Jacobian. Equation (32) shows this for a general regular observation family, with the Gamma density kg_W^2/μ^2 as a specialization. A microscopic speckle construction preserves the qualitative geometry while modestly changing its response-level weighting.

The calibration geometry contains an exact scale orbit: when c and linear modal amplitudes are both unknown, only the products ca_i are identified by the Bessel observation. The numerical studies reproduce that null, verify the analytic gradient to machine precision, and show how the local theory behaves as the inverse problem becomes harder. Modal constraints retain accurate field recovery across several pointwise Bessel branches; in regular first-branch recovery, $\Phi F_{\boldsymbol{a}}^{-1}\Phi^T$ predicts the spatial

uncertainty field, whereas exact stationary responses instantiate the classical nonregular quarter-power regime specialized in Proposition 4 and observed numerically with approximately $N^{-1/4}$ error scaling.

A phase-referenced vibration channel completes the residual structural step. With a fixed external phase origin and nonzero sampled amplitude, the same quantity Q controls oracle sign separation, phase information, curvature along the MESPI scale null, and curvature of the joint scale ridge. A balanced oracle Monte Carlo confirms $\Phi(-\sqrt{Q})$ for the simple-hypothesis sign problem, including the declared 10^{-3} design cell at $Q^* = 9.5495$. A separate operational profile-MLE test, in which the optical magnitude representative is estimated first, shows monotonically decreasing sign error with Q but a statistically resolved mid-Q departure from the oracle law; that departure is reported without causal attribution. Conditional on successful magnitude-branch localization, increasing Q also contracts metric-scale uncertainty. A paired decomposition using the same stored optical estimate in each realization isolates an auxiliary contribution whose high-Q standard deviation scales as $Q^{-0.507}$ ($R^2 = 0.9999$), consistent with the local $Q^{-1/2}$ curvature prediction. At the nominal frozen operating point the total log-scale SD is already within 1.55% of the deterministic zero-auxiliary-noise optical floor, so further auxiliary-noise reduction yields little scale-precision gain unless the preceding optical magnitude estimate is improved. This floor is specific to the tested operational ensemble. The nodal fraction 3.92% remains an oracle auxiliary-channel exclusion threshold rather than an end-to-end pipeline guarantee. The resulting chain

$$2^{N_{\text{nodal}}} \longrightarrow 2 \longrightarrow 1 \tag{78}$$

organizes the inverse problem by source of information: optical symmetry sets the initial equivalence class, mechanics contracts it, and an independent phase/scale reference selects the metric signed field. This structure gives a clear experimental program for calibrated time-averaged ESPI inversion and a well-defined differentiable measurement layer for subsequent physics-guided estimators.

A Reproducibility details

The v16.5 release includes manuscript-level traceability files, the unified C1/C2/F2/F3 study package, the independently audited frozen Study G v2 package, the Study F4 field-covariance package, and a post-freeze Study-G3 extension that imports the frozen Study-F basis and sampling configuration without modifying the audited G0–G3 evidence. These sources retain numerical configuration, code, machine-readable results, validation records, fixed seeds, and the resource amendments recorded before the final F2/F3 and Study G outcomes. Numerical claims used in the manuscript are mapped to source artifacts and keys by a traceability table provided as Table S1 of the Supplement.

Study C1 is deterministic and records the six-domain adjacency graph, exact plate energy, grid-refinement sequence, and all 64 optical sign assignments. Study C2 records the microscopic speckle moments, Monte Carlo checks of the sample-variance mean and variance, and the variance-aware weighting analysis. The F2/F3 package contains the multibranch Sobol audit, paired Gamma Monte Carlo runs, predictive CRLB experiment, and stationary scaling study; a resource-driven reduction of the planned F2/F3 Monte Carlo and multistart counts was recorded before the final outcomes in the package amendment log. Study F4 is packaged separately with its fixed seed, 1,000-run per-k modal ensembles, predicted and empirical covariance arrays, acceptance gates, and deterministic figure source. Study G uses a frozen numerical configuration, recorded seeds, and command history. The v16.1 extension records the frozen configuration and source hashes, performs balanced 100,000-trial oracle calibration, including the declared $Q^* = 9.5495$ design cell, stores the 1,000-realization operational profile-MLE outputs separately, and adds an extended-Q scale-profile sweep that reuses the frozen optical fits without rerunning the optical inversion. The v16.4 paired diagnostic reuses those same stored optical estimates and child seeds, computes $\Delta\xi_Q$ relative to each realization's deterministic zero-auxiliary-noise profile, and adds the nominal $Q = 1.7657 \times 10^4$ auxiliary-noise evaluation without any optical refit. Derived members of the central Study-G identity remain printed symbolically as $2Q$ and $4Q$, preventing independent rounding from breaking the exact relation in Eq. (61).

The main manuscript uses reproducible figures from these sources, `generate_revision_diagnostics.py`, the analytical operator-geometry generator, or the Study F4 generator. Exploratory Fisher-design and local-CRLB figures without a complete generator path remain excluded from the current evidence set.

B Experimental requirements for MESPI inversion

Table 3 summarizes the data blocks required for a calibrated inversion study.

Table 3: Compact calibration checklist for MESPI inversion.

Record group	Required controls and purpose
D0–**D1** Optical calibration	Exposure/gain, bit depth, dark/linearity/saturation checks, wavelength, geometry, spatial scale, and A, B, c; establishes the quantitative observable and optical scale.
D2, D6 Branch control	Registered amplitude sweep across relevant response levels plus fringe-order anchors or multifrequency tracking; tests the J_0^2 profile and selects the magnitude branch.
D3–**D4** Phase / polarity	Synchronized SLDV/LDV with phase origin known modulo 2π, documented excitation–response polarity, and response-positive direction relative to the optical sensitivity vector; validates scale and fixes absolute sign.
D5 Stochastic calibration	Repeated acquisitions with timing/telemetry for stationarity, spatial dependence, and $k(x)$ or alternative-noise fitting; calibrates the likelihood and empirical uncertainty.

The sequence matters operationally. Radiometric and optical calibration establish the response scale; repeated acquisitions establish the likelihood; branch information identifies the relevant local inverse; and the externally referenced vibration channel supplies absolute sign and displacement scale. Repeated inverse estimates can then be compared with the Fisher/CRLB prediction at regular operating points and with the nonregular behavior expected near stationary response levels.

Disclosures

The author declares no conflicts of interest.

Data availability

The numerical evidence is fully synthetic and generated procedurally. The v16.5 release contains the manuscript and supplement sources, figures, numerical traceability files, the unified C1/C2/F2/F3 archive, the independently audited frozen

Study G v2 package, the fixed-seed Study F4 field-covariance package, and the post-freeze Study-G3 oracle/operational sign-and-scale extension, including the Q^* calibration cell, extended-Q scale sweep, deterministic zero-auxiliary-noise scale limit, paired auxiliary-scale diagnostic at the nominal operating point, machine-readable outputs, and source hashes.

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